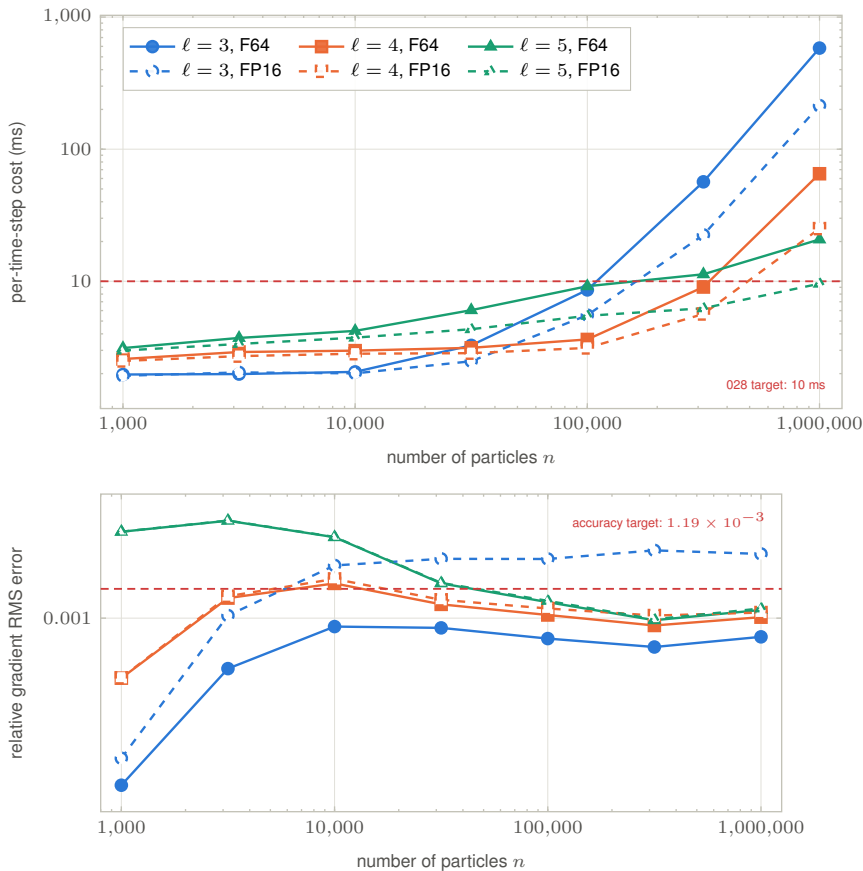


Fig. 10 — Per-step cost and accuracy vs n at the task-028 defaults, fixed depth



Regime: $n \in \{1000, 3162, 10^4, 31623, 10^5, 316228, 10^6\}$, literature $P = 4$ (code `expansion_order=3`), one H200, seed 24025, fixed box $(-0.01, 1.02)$ so the grid is identical at every n . Task-028 shipped defaults: hierarchical rigid stencil, dense M2L, one route window per level, bounded Morton counting sort, 64 threads / 65 536 block cap. **Boundary:** one complete recurring resident step — tree/route refresh, evaluation (B2M $\rightarrow$ M2M $\rightarrow$ M2L $\rightarrow$ L2L $\rightarrow$ L2B + nearfield), output finalization and device Euler convection — with zero per-step body transfers, enforced by an asserted counter contract. Median over 9 repetitions. **Series:** three *fixed* depths $\ell = 3/4/5$, each with the shipped $(6, 5, \dots, 5)$ level radius schedule, held constant across the whole sweep; solid = Float64, dashed/hollow = the FP16-WMMA/Float32 winner configuration. **Reading the lower panel:** the geometry was tuned at $n = 10^6$, where $\ell = 5/\text{FP16}$ delivers 1.0593×10^{-3} against the 1.19×10^{-3} gate. At other n the same fixed geometry delivers a different accuracy; that drift is the measured quantity motivating per- n retuning, not a defect. No pass/fail error gate is applied to these points. Errors are sampled-direct against the checksummed 024b references (512 bodies, seeds 24025/24026). **Data:** task 030, `data/cost_vs_n/cuda030_*.csv`.